

PROJECT REPORT

ON

Incremental Open-Set Recognition Using DINOv2 for Aerial Scene Understanding

Submitted by

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ABOUT ADRDE

Aerial Delivery Research and Development Establishment (ADRDE), is a pioneer establishment in the country engaged in design and development of flexible aerodynamics decelerates commonly known as parachutes and the allied technologies. This establishment has been recognized to improve its technical and administrative efficiency as well as to increase the work output and to have better interaction between the scientific communities of this establishment. Four technology groups in the field of aerodynamics, mechanical, instrumentation and textile for assistance in design support their project divisions. ADRDE is an ISO 9001:2015 certified lab. This establishment has been responding well to meet the ever-increasing requirements of the Services & Space organization by developing state-of-the-art systems. An independent information technology division has also been treated to control and monitor the program on the project as well as to carry out micro and macro planning, to meet the user requirements effectively.

CERTIFICATE

This is to certify that the project report entitled “**Incremental Open-Set Recognition Using DINOv2 for Aerial Scene Understanding**” submitted by **Tanya Prabhakar (23UMATH027)** to the School of Physical Sciences, **Central University of Karnataka**, Kalaburagi, in partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology in Mathematics and Computing**, is a bona fide record of the work carried out by the student under the supervision and guidance of the undersigned during the academic year 2025–26.

It is further certified that this work has not been submitted, either in whole or in part, to any other university or institution for the award of any degree, diploma, or other academic qualification. The project embodies the original work of the student, and all sources of information used in the preparation of this report have been duly acknowledged.

The work presented in this report is found to be satisfactory and worthy of submission for evaluation.

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DECLARATION

I hereby declare that the project report entitled "**Incremental Open-Set Recognition Using DINOv2 for Aerial Scene Understanding**" is a record of the work carried out by me during my 45-day internship at **Aerial Delivery Research and Development Establishment (ADRDE), DRDO**, from 11 May 2026 to 24 June 2026, under the guidance of **Mr. Ajay Kumar, Scientist 'B'**, and review of **Mr. Ajitabh Tiwari, Scientist 'E'**.

I declare that the work presented in this report is my own and has been completed as part of the internship requirements. Wherever the work of others has been used, appropriate acknowledgement and references have been provided.

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ABSTRACT

Remote sensing image retrieval in open-world settings presents a fundamental challenge: systems deployed in the real world will inevitably encounter image categories never seen during training and must handle them gracefully rather than forcing incorrect predictions. This work presents a framework for open-world continual semantic retrieval over remote sensing imagery addressing three interconnected problems, retrieving visually similar images from a known gallery, detecting images from unknown categories, and incorporating new categories at runtime without retraining.

The framework is built around a frozen DINOv2 Vision Transformer backbone, providing rich, generalizable visual representations without representation drift during continual updates. Features are extracted from multiple transformer layers and combined with attention-pooled patch representations to capture both global scene semantics and local discriminative detail. A projection head and attention pooling module are trained using supervised contrastive loss with structured PK sampling, organizing the embedding space so that same-class images cluster tightly and different-class images remain well separated. All embeddings are L2-normalized and indexed using FAISS for exact cosine similarity search.

Open-set recognition is handled through per-class adaptive similarity thresholds, each calibrated from the intra-class similarity distribution of the corresponding gallery class. Queries falling below the class threshold are classified as unknown, avoiding the limitations of a single global threshold that cannot account for variation in class compactness.

Continual class registration is achieved by appending new class embeddings directly to the FAISS index without modifying any model weights, eliminating catastrophic forgetting by design. The framework is evaluated on a subset of NWPU-RESISC45 comprising 15 known and 3 unknown classes. Continual registration of all three previously unknown classes yields Recall@5 of up to 1.00 and mAP of up to 0.99, demonstrating effective generalization to new categories without retraining. The results confirm that treating open-world recognition as a retrieval problem, grounded in stable pre-trained representations and calibrated similarity thresholds, is a practical approach for real-world remote sensing applications.

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